1 Scale factor $= \dfrac{6+3}{6} = 1.5$ (or ratio $\dfrac{QR}{8} = \dfrac{9}{6}$) **[M1]**

12 **[A1]**

[Total Marks: 2]

2 **(a)** Finds scale factor: $\dfrac{4+6}{4} = 2.5$ **[M1]**

12.5 **[A1]**

(b) Uses area scale factor: $16 \times (2.5)^2$ **[M1]**

100 **[A1]**

[Total Marks: 4]

3 **(a)** Linear scale factor $= \dfrac{4+2}{4} = 1.5$ (or equivalent ratio $\dfrac{6}{4}$) **[M1]**

7.5 **[A1]**

(b) Area of $ABC = 8 \times 1.5^2 = 18$ **[M1]**

10 **[A1]**

[Total Marks: 4]

4 **(a)** Scale factor $= \dfrac{18}{6} = 3$ **[M1]**

15 **[A1]**

(b) Area scale factor $= 3^2 = 9$ **[M1]**

$\dfrac{72}{9} = 8$ **[A1]**

[Total Marks: 4]

5 **(a)** Volume scale factor $= \left(\dfrac{15}{6}\right)^3 = \dfrac{125}{8}$ **[M1]**

$V_B = 48 \times \dfrac{125}{8} = 750$ **[A1]**

(b) Area scale factor $= \left(\dfrac{15}{6}\right)^2 = \dfrac{25}{4}$ **[M1]**

$A_A = 200 \div \dfrac{25}{4} = 32$ **[A1]**

[Total Marks: 4]

6 **(a)** Length scale factor $= \dfrac{9}{3} = 3$ **[M1]**

15 **[A1]**

(b) Area scale factor $= 3^2 = 9$ **[M1]**

6 **[A1]**

[Total Marks: 4]

7 (a) Scale factor $= \dfrac{4+2}{4} = 1.5$ (or $\dfrac{6}{4}$ or $\dfrac{3}{2}$) **[M1]**

7.5 **[A1]**

(b) Area scale factor $= 1.5^2 = 2.25$ **[M1]**

Area of triangle $ABC = 8 \times 2.25 = 18$ **[M1]**

10 **[A1]**

[Total Marks: 5]

8 (a) Scale factor $= \dfrac{4+2}{4}$ or 1.5 **[M1]**

9 **[A1]**

(b) Area scale factor $= 1.5^2$ or 2.25 **[M1]**

Area of triangle $ABC = 32 \times 2.25$ or 72 **[M1]**

40 **[A1]**

[Total Marks: 5]

9 (a) Identifies the scale factor is $\dfrac{AD+DB}{AD} = \dfrac{6}{4}$ or 1.5 **[M1]**

9 **[A1]**

(b) Uses the area scale factor: 1.5^2 or $\dfrac{9}{4}$ or 2.25 **[M1]**

Calculates the area of triangle ABC: $12 \times 2.25 = 27$ **[M1]**

15 **[A1]**

[Total Marks: 5]

10 (a) Finds the linear scale factor: $\dfrac{15}{10} = 1.5$ (or $\dfrac{3}{2}$) **[M1]**

Finds the area scale factor: $1.5^2 = 2.25$ (or $\dfrac{9}{4}$) **[M1]**

$120 \times 2.25 = 270$ **[A1]**

(b) Finds the volume scale factor: $1.5^3 = 3.375$ (or $\dfrac{27}{8}$) **[M1]**

Sets up the correct division: $540 \div 3.375$ (or $540 \times \dfrac{8}{27}$) **[M1]**

160 **[A1]**

[Total Marks: 6]

11 **(a)** $\frac{EF}{6} = \frac{10}{4}$ (or scale factor = 2.5) **[M1]**

15 **[A1]**

(b) $AC = 12.5 \times \frac{4}{10}$ (or $12.5 \div 2.5$) **[M1]**

5 **[A1]**

(c) Area scale factor $= \left(\frac{10}{4}\right)^2 = 6.25$ (or $\frac{25}{4}$) **[M1]**

$10 \times 6.25 = 62.5$ **[A1]**

[Total Marks: 6]

12 **(a)** $\frac{6}{9} = \frac{8}{QR}$ or scale factor $= \frac{9}{6}$ (or 1.5) **[M1]**

12 **[A1]**

(b) $\frac{6}{9} = \frac{PT}{12}$ or $PT = 12 \div 1.5$ **[M1]**

8 **[A1]**

(c) Area scale factor $= \left(\frac{9}{6}\right)^2$ or 1.5^2 or 2.25 **[M1]**

Area of triangle $PQR = 20 \times 2.25 = 45$ **[M1]**

25 **[A1]**

[Total Marks: 7]

13 **(a)** Area scale factor $= \frac{45}{20}$ or $\frac{9}{4}$ **[M1]**

Linear scale factor $= \sqrt{\frac{9}{4}} = \frac{3}{2}$ **[M1]**

Volume $B = 16 \times \left(\frac{3}{2}\right)^3 = 16 \times \frac{27}{8} = 54$ **[A1]**

(b) Height $A = 12 \div \frac{3}{2}$ or $12 \times \frac{2}{3}$ **[M1]**

8 **[A1]**

[Total Marks: 5]

14 **(a)** Volume scale factor $= \frac{1350}{400} = \frac{27}{8}$ (or 3.375) **[M1]**

Linear scale factor $= \sqrt[3]{\frac{27}{8}} = \frac{3}{2}$ (or 1.5) **[M1]**

18 **[A1]**

(b) Area scale factor $= \left(\frac{3}{2}\right)^2 = \frac{9}{4}$ (or 2.25) **[M1]**

$\dfrac{360}{2.25} = 160$ **[A1]**

[Total Marks: 5]

15 (a) Angle AEB = angle CED (vertically opposite angles) **[B1]**

Angle BAE = angle DCE (alternate angles, AB parallel to DC) **[B1]**

Angle ABE = angle CDE (alternate angles, AB parallel to DC), so all three angles are equal **[B1]**

(b) (i) $\dfrac{EC}{4} = \dfrac{9}{6}$ **[M1]**

6 **[A1]**

(ii) $\dfrac{BE}{12.5 - BE} = \dfrac{6}{9}$ (or $BE = \dfrac{2}{5} \times 12.5$) **[M1]**

5 **[A1]**

(c) $12 \times \left(\dfrac{9}{6}\right)^2$ **[M1]**

27 **[A1]**

[Total Marks: 9]

16 (a) Angle A (or $\angle DAE$ and $\angle CAB$) is common **[B1]**

Angle ADE = Angle ACB (given) **[B1]**

Therefore the third angles are equal ($\angle AED = \angle ABC$) and the triangles are similar (AAA) **[B1]**

(b) Sets up correct ratio: $\dfrac{AD}{AC} = \dfrac{AE}{AB}$ or $\dfrac{x}{4+5} = \dfrac{4}{x+(x+1)}$ **[M1]**

$\dfrac{x}{9} = \dfrac{4}{2x+1}$ **[M1]**

$2x^2 + x - 36 = 0$ **[M1]**

$(2x+9)(x-4) = 0$ implies $x = 4$ (ignoring negative root) **[A1]**

(c) Area scale factor $= \left(\dfrac{4}{9}\right)^2 = \dfrac{16}{81}$ **[M1]**

16 : 65 **[A1]**

[Total Marks: 9]

17 (a) Linear scale factor $= 14 \div 4$ or 3.5 **[M1]**

21 **[A1]**

(b) Area scale factor $= 3.5^2$ or 12.25 **[M1]**

104.125 **[A1]**

[Total Marks: 4]

18 **(a)** Area scale factor $= \frac{800}{450}$ (or $\frac{16}{9}$) **[M1]**

Length scale factor $= \sqrt{\frac{16}{9}} = \frac{4}{3}$ and Volume scale factor $= \left(\frac{4}{3}\right)^3 = \frac{64}{27}$ **[M1]**

3200 **[A1]**

(b) Uses length scale factor: $48 \div \frac{4}{3}$ (or $48 \times \frac{3}{4}$) **[M1]**

36 **[A1]**

[Total Marks: 5]

19 **(a)** Volume scale factor $= \frac{1200}{150} = 8$ **[M1]**

Length scale factor $= \sqrt[3]{8} = 2$ **[M1]**

24 **[A1]**

(b) Area scale factor $= 2^2 = 4$ **[M1]**

110 **[A1]**

[Total Marks: 5]

20 **(a)** Volume scale factor $= \frac{1080}{320}$ or $\frac{27}{8}$ or 3.375 **[M1]**

Linear scale factor $= \sqrt[3]{\frac{27}{8}}$ or 1.5 **[M1]**

27 **[A1]**

(b) Area scale factor $= 1.5^2$ or 2.25 or $\frac{9}{4}$ **[M1]**

Surface area of $A = 630 \div 2.25$ or $630 \times \frac{4}{9}$ **[M1]**

280 **[A1]**

[Total Marks: 6]

21 **(a)** Volume scale factor $= \left(\frac{30}{12}\right)^3$ or 2.5^3 or 15.625 **[M1]**

5000 **[A1]**

(b) Area scale factor $= \left(\frac{30}{12}\right)^2$ or 2.5^2 or 6.25 **[M1]**

$1125 \div 6.25 = 180$ **[A1]**

(c) $4.80 \times \left(\frac{30}{12}\right)^3$ or 4.80×15.625 **[M1]**

75 or 75.00 **[A1]**

[Total Marks: 6]

22 **(a)** Volume scale factor $= \dfrac{8640}{320} = 27$ **[M1]**

Length scale factor $= \sqrt[3]{27} = 3$ **[M1]**

36 **[A1]**

(b) Area scale factor $= 3^2 = 9$ **[M1]**

250 **[A1]**

(c) Mass $= 8.5 \times 320 = 2720\,\text{g}$ **[M1]**

2.72 **[A1]**

[Total Marks: 7]

23 **(a)** Angle A is common to both triangles **[B1]**

Angle ABC = Angle ADE (corresponding angles as BC is parallel to DE) **[B1]**

(b) $\dfrac{DE}{7.5} = \dfrac{6+4}{6}$ **[M1]**

12.5 **[A1]**

(c) Area scale factor $= \left(\dfrac{10}{6}\right)^2 = \dfrac{25}{9}$ **[M1]**

Area of $ADE = 18 \times \dfrac{25}{9} = 50$ **[M1]**

Area of $BCED = 50 - 18 = 32$ **[A1]**

[Total Marks: 7]

24 **(a)** States that angle A is common to both triangles. **[B1]**

States that angle ADE = angle ABC (and/or angle AED = angle ACB) because they are corresponding angles (due to parallel lines). **[B1]**

(b) Calculates the full length of AB: $6 + 4 = 10$ **[M1]**

Finds the linear scale factor: $\dfrac{10}{6} = \dfrac{5}{3}$ and sets up equation $BC = 9 \times \dfrac{5}{3}$ **[M1]**

15 **[A1]**

(c) Finds the area scale factor: $\left(\dfrac{5}{3}\right)^2 = \dfrac{25}{9}$ **[M1]**

Calculates the area of triangle ABC: $27 \times \dfrac{25}{9} = 75$ **[M1]**

Calculates the area of the trapezium: $75 - 27 = 48$ **[A1]**

[Total Marks: 8]

25 **(a)** Linear scale factor $= \dfrac{45}{15} = 3$ **[M1]**

Mass (Volume) scale factor $= 3^3 = 27$ **[M1]**

13.5 **[A1]**

(b) Mass scale factor from P to $R = \dfrac{4000}{500} = 8$ **[M1]**

Linear scale factor $= \sqrt[3]{8} = 2$ **[M1]**

30 **[A1]**

(c) Area scale factor from P to $Q = 3^2 = 9$ **[M1]**

270 **[A1]**

[Total Marks: 8]

26 (a) Calculates the volume ratio: $\dfrac{1080}{320} = \dfrac{27}{8}$ or 3.375 **[M1]**

Calculates the length scale factor: $\sqrt[3]{\dfrac{27}{8}} = \dfrac{3}{2}$ or 1.5 **[M1]**

18 **[A1]**

(b) Uses the area scale factor: $450 \div 1.5^2$ or $450 \times \left(\dfrac{2}{3}\right)^2$ **[M1]**

200 **[A1]**

(c) Calculates the volume of cone R: $320 + 1080 = 1400$ **[B1]**

Finds the length scale factor from P to R: $\sqrt[3]{\dfrac{1400}{320}}$ or $\sqrt[3]{4.375}$ **[M1]**

Calculates the height of R: $12 \times \sqrt[3]{4.375} = 19.626\ldots$ and rounds to 19.6 **[A1]**

[Total Marks: 8]

27 (a) Angle AXB = Angle CXD (vertically opposite) **[B1]**

Angle ABX = Angle CDX (alternate angles, AB parallel to DC) **[B1]**

(Accept Angle BAX = Angle DCX for the second B1 mark)

(b) Length scale factor $= \dfrac{15}{9} = \dfrac{5}{3}$ **[M1]**

$DX = 6 \times \dfrac{5}{3} = 10$ **[M1]**

$BD = 6 + 10 = 16$ **[A1]**

(c) Area scale factor $= \left(\dfrac{15}{9}\right)^2 = \dfrac{25}{9}$ **[M1]**

Area $= 27 \times \dfrac{25}{9} = 75$ **[A1]**

(d) Ratio of bases $\dfrac{CX}{AX} = \dfrac{15}{9} = \dfrac{5}{3}$ **[M1]**

Area $= 27 \times \dfrac{5}{3} = 45$ **[A1]**

[Total Marks: 9]

28 **(a)** $\dfrac{4}{4+x} = \dfrac{5}{7+x}$ OR $\dfrac{4}{x} = \dfrac{5}{x+2}$ **[M1]**

$4(x+7) = 5(x+4)$ OR $4(x+2) = 5x$ **[M1]**

$x = 8$ **[A1]**

(b) Linear scale factor $= \dfrac{4+8}{4} = 3$ **[M1]**

$BC = 3.5 \times 3 = 10.5$ **[A1]**

(c) Area of $\triangle ABC = 3^2 \times A = 9A$ **[M1]**

Area of trapezium $= 9A - A = 8A$ **[A1]**

[Total Marks: 7]

29 **(a)** Volume ratio $A : B = \dfrac{108}{500} = \dfrac{27}{125}$ **[M1]**

Area ratio $= \left(\sqrt[3]{\dfrac{27}{125}}\right)^2 = \left(\dfrac{3}{5}\right)^2 = \dfrac{9}{25}$ **[M1]**

Let surface areas be $9x$ and $25x$. $9x + 25x = 340 \implies 34x = 340 \implies x = 10$ **[M1]**

$25 \times 10 = 250$ **[A1]**

(b) Volume of small cone $= 1 - \dfrac{7}{8} = \dfrac{1}{8}$ of Cone B **[M1]**

Ratio of heights $= \sqrt[3]{\dfrac{1}{8}} = \dfrac{1}{2}$ **[M1]**

Height of small cone $= 20 \times \dfrac{1}{2} = 10$ **[A1]**

[Total Marks: 7]

30 **(a)** $\angle AXB = \angle DXC$ (vertically opposite angles) **[B1]**

$\angle BAX = \angle CDX$ OR $\angle ABX = \angle DCX$ (angles in the same segment) **[B1]**

Third angle in triangle OR states the other pair of angles in the same segment, concluding triangles are similar (AAA) **[B1]**

(b) Linear scale factor $= \dfrac{7.5}{5} = 1.5$ OR $\dfrac{DX}{4} = \dfrac{7.5}{5}$ **[M1]**

$DX = 6$ **[A1]**

$DC = 9$ **[A1]**

(c) Area scale factor $= 1.5^2 = 2.25$ **[M1]**

26.55 **[A1]**

[Total Marks: 8]

31 **(a)** $\angle AXB = \angle DXC$ because they are vertically opposite angles **[B1]**

$\angle BAX = \angle CDX$ (or $\angle ABX = \angle DCX$) because they are angles subtended by the same arc in the same segment **[B1]**

Therefore, the triangles are similar by AAA similarity **[B1]**

(b) Uses property of intersecting chords (or similar triangle ratios):
$(x+2)(x-2) = (x-3)(x+4)$ **[M1]**

Expands brackets correctly: $x^2 - 4 = x^2 + x - 12$ **[M1]**

8 **[A1]**

(c) Calculates side lengths $AX = 10$ and $DX = 12$, and sets up area scale factor: $45 \times \left(\frac{10}{12}\right)^2$ **[M1]**

31.25 **[A1]**

[Total Marks: 8]

32 (a) $\frac{x+2}{(x+2)+(x-1)} = \frac{8}{8+x}$ or $\frac{x+2}{x-1} = \frac{8}{x}$ **[M1]**

$(x+2)(x+8) = 8(2x+1)$ or $x(x+2) = 8(x-1)$ **[M1]**

$x^2 + 10x + 16 = 16x + 8$ or $x^2 + 2x = 8x - 8$ leading to $x^2 - 6x + 8 = 0$ **[A1]**

(b) $(x-2)(x-4) = 0$ **[M1]**

2 and 4 **[A1]**

(c) When $x = 4$, length ratio $\frac{AD}{AB} = \frac{6}{9} = \frac{2}{3}$ **[M1]**

Area of $ABC = 24 \div \left(\frac{2}{3}\right)^2 = 54$ **[M1]**

30 **[A1]**

[Total Marks: 8]

33 (a) Angle A is common to both triangles **[B1]**

Angle AED = Angle ABC (given in the question) **[B1]**

Therefore, Angle ADE = Angle ACB (sum of angles in a triangle is 180°), so the triangles are similar (AAA) **[B1]**

(b) $AB = 6 + 9 = 15$ cm **[M1]**

$\frac{AC}{AD} = \frac{AB}{AE} \implies \frac{AC}{6} = \frac{15}{5}$ **[M1]**

$AC = 18$, so $EC = 18 - 5 = 13$ **[A1]**

(c) Area scale factor for the triangles $= \left(\frac{5}{15}\right)^2 = \frac{1}{9}$ **[M1]**

1 : 8 **[A1]**

[Total Marks: 8]

34 (a) States that angle BAX = angle DCX OR angle ABX = angle CDX due to alternate angles (between parallel lines) **[B1]**

States a second valid pair of angles are equal (e.g. vertically opposite angles at X or the other pair of alternate angles) concluding the triangles are similar (AAA) **[B1]**

(b) Identifies the length scale factor: $\frac{10.5}{7}$ or 1.5 **[M1]**

Calculates $CX = 4 \times 1.5 = 6$ **[M1]**

10 **[A1]**

(c) Calculates the area of triangle CDX: $16 \times 1.5^2 = 36$ **[M1]**

Recognises the ratio of areas of triangle BCX to triangle ABX is equal to the ratio of bases $CX : AX = 1.5 : 1$ (as they share the same height from B) **[M1]**

Calculates the area of triangle BCX (or ADX): $16 \times 1.5 = 24$ **[M1]**

Total area $= 16 + 36 + 24 + 24 = 100$ **[A1]**

[Total Marks: 9]

35 (a) Angle A is common to both triangles **[B1]**

Angle ABC = Angle AED (given), so the third angles must be equal, hence AA similarity **[B1]**

(b) 2.5 or $\frac{5}{2}$ **[B1]**

(c) $\frac{AB}{AE} = 2.5 \implies \frac{2x + 2.5}{x} = 2.5$ (or equivalent valid ratio) **[M1]**

$2x + 2.5 = 2.5x$ **[M1]**

$x = 5$ **[A1]**

(d) Area scale factor $= 2.5^2 = 6.25$ **[M1]**

Formulates correct equation for areas, e.g., $6.25A - A = 42$ or $5.25A = 42$ **[M1]**

8 **[A1]**

[Total Marks: 9]

36 (a) $\angle AXB = \angle CXD$ because they are vertically opposite angles **[B1]**

$\angle BAX = \angle DCX$ (or $\angle ABX = \angle CDX$) because they are alternate angles, $AB \parallel DC$ **[B1]**

All three pairs of corresponding angles are equal, so the triangles are similar (AAA) **[B1]**

(b) $\frac{2x - 3}{x} = \frac{6}{4}$ or equivalent **[M1]**

$4(2x - 3) = 6x$ or $2x - 3 = 1.5x$ **[M1]**

$0.5x = 3$ leading to $x = 6$ **[A1]**

(c) Area of triangle $CDX = 24 \times 1.5^2$ or 54 **[M1]**

Recognises triangle BCX shares altitude with ABX and base ratio is 6 : 4 **[M1]**

Area of triangle BCX (and triangle ADX) $= 24 \times 1.5$ or 36 **[M1]**

Total area $= 24 + 54 + 36 + 36 = 150$ **[A1]**

[Total Marks: 10]

37 **(a)** Finds the volume scale factor: $\frac{2700}{800} = 3.375$ (or $\frac{27}{8}$) **[M1]**

Finds the linear scale factor by cube rooting: $\sqrt[3]{3.375} = 1.5$ **[M1]**

$12 \times 1.5 = 18$ **[A1]**

(b) Finds the area scale factor from the linear scale factor: $1.5^2 = 2.25$ **[M1]**

Sets up the multiplication for the cost: 18×2.25 **[M1]**

40.50 **[A1]**

(c) Identifies the area scale factor between C and A is 1.44 **[B1]**

Finds the linear scale factor: $\sqrt{1.44} = 1.2$ **[M1]**

Finds the volume scale factor: $1.2^3 = 1.728$ **[M1]**

Calculates the volume of C: $800 \times 1.728 = 1382.4$ **[A1]**

[Total Marks: 10]

38 **(a)** Slant height $l = \sqrt{10^2 + 24^2} = 26$ **[M1]**

Surface Area $= \pi(10)^2 + \pi(10)(26)$ **[M1]**

360π **[A1]**

(b) $V_A = \frac{1}{3}\pi(10)^2(24) = 800\pi$ **[M1]**

Volume Ratio $= \frac{100\pi}{800\pi} = \frac{1}{8}$ **[M1]**

Area Ratio $= \left(\frac{1}{8}\right)^{\frac{2}{3}} = \frac{1}{4}$ **[M1]**

Surface Area of $B = 360\pi \times \frac{1}{4} = 90\pi$ **[A1]**

(c) Volume of frustum $= 800\pi - 100\pi = 700\pi$ **[M1]**

$\frac{4}{3}\pi R^3 = 700\pi$ **[M1]**

$R^3 = 700 \times \frac{3}{4} = 525$ **[M1]**

$R = \sqrt[3]{525} = 8.07$ (accept $8.066\ldots$) **[A1]**

[Total Marks: 11]

39 **(a)** $320 \times \left(\frac{24}{15}\right)^2$ **[M1]**

819.2 **[A1]**

(b) $\frac{V_X}{2048} = \left(\frac{15}{24}\right)^3$ **[M1]**

Volume scale factor $= 1.6^3 = 4.096$ (or $\frac{512}{125}$) **[M1]**

500 **[A1]**

(c) $\dfrac{h_Z}{15} = \sqrt[3]{\dfrac{1372}{500}}$ **[M1]**

Scale factor $= \sqrt[3]{2.744} = 1.4$ **[A1]**

$15 \times 1.4 = 21$ **[A1]**

(d) Number of complete copies $= 4$ (from $\dfrac{2048}{500} = 4.096$) **[M1]**

Remaining volume $= 2048 - 4 \times 500 = 48$ cm^3 **[A1]**

$\dfrac{4}{3}\pi r^3 = 48$ giving $r^3 = \dfrac{36}{\pi}$ $(= 11.459\ldots)$ **[M1]**

2.25 **[A1]**

[Total Marks: 12]